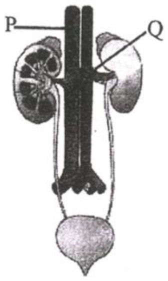
Excretory Products And Their Elimination

1. **Juxta glomerular apparatus in humans is formed by the cellular modifications in the**
   (1) PCT and efferent arteriole at the location of their contact
   (2) DCT and efferent arteriole at the location of their contact
   (3) DCT and afferent arteriole at the location of their contact
   (4) PCT and afferent arteriole at the location of their contact
2. **Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).**
   Assertion(A): Ammonia should be removed from the body as rapidly as it is formed.
   Reason(R): In water, ammonia is insoluble.
   In the light of the above statements, choose the most appropriate answer from option given below
   (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
   (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
   (3) Assertion (A) is true, and Reason (R) is false.
   (4) Assertion (A) is false, and Reason (R) is true.
3. **Proximal convoluted tubule (PCT) is**
   (1) Simple cuboidal epithelium with brush border
   (2) Simple cuboidal epithelium without brush border
   (3) Simple columnar epithelium with brush border
   (4) Simple columnar epithelium without brush border
4. **During the process of urine formation in medullary region of kidney, minimum reabsorption occurs in**
   (1) PCT
   (2) Descending limb of loop of Henle
   (3) Ascending limb of loop of Henle
   (4) Distal convoluted tubule
5. **Which of the following is incorrect about human kidney?**
   (1) Kidney is covered by tough capsule.
   (2) Kidney is divided into inner cortex and medulla on outer side.
   (3) The cortex extended in between the medullary pyramid are renal column of Bertini.
   (4) Kidney is situated close to the dorsal inner wall of abdominal cavity.
6. **Match the column.**

Column-I	Column-II
A. Flame cells	**1. Prawn**
B. Malpighian tubules	**2. Cockroach**
C. Nephridia	**3. Planaria**
D. Green gland	**4. Earthworm**

   Choose the most appropriate answer from option given below
   (1) A-3, B-2, C-4, D-1
   (2) A-2, B-3, C-1, D-4
   (3) A-3, B-4, C-2, D-1
   (4) A-4, B-2, C-3, D-1
7. **Which of the following is not the process involved in urine formation?**
   (1) Glomerular filtration
   (2) Micturition
   (3) Secretion
   (4) Reabsorption

8. **Select the option with correct identification in human excretory system?**



P	Q
(1) Dorsal artery	Renal vein
(2) Dorsal artery	Renal artery
(3) Inferior vena cava	Renal vein
(4) Inferior vena cava	Renal artery

9. **Which of the following statements is not true w.r.t. different types of nephrons in human body?**

(1) Cortical nephrons are more abundant as compared to juxtamedullary nephrons

(2) Vasa recta is reduced or absent in cortical nephrons

(3) Juxtamedullary nephrons have longer loop of Henle as compared to cortical nephrons that runs deep into the medulla

(4) The Malpighian corpuscle, PCT and DCT are present in the medullary region of the kidney

10. **Given below are two statements:**

Statement-I: Inner to the hilum is a broad funnel shaped space called the renal pelvis with projections called calyces.

Statement-II: The inner layer of kidney is a tough capsule.

Choose the most appropriate answer from option given below

(1) Statement I and Statement II both are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

11. **Glomerulus is a tuft of capillaries formed by _____(a)_____. Blood from the glomerulus is carried away by an _____(b)_____.**

(1) (a) – efferent arteriole; (b) – afferent arteriole

(2) (a) – efferent arteriole; (b) – efferent arteriole

(3) (a) – afferent arteriole; (b) – afferent arteriole

(4) (a) – afferent arteriole; (b) – efferent arteriole

12. **The amount of glomerular filtrate formed by the kidneys in 10 hr in a healthy human is approximately**

(1) 45 litres

(2) 75 litres

(3) 95 litres

(4) 180 litres

13. **Match the column-I with column-II**

	Column-I		Column-II
I.	**Proximal convoluted tubule**	**A.**	**70-80% reabsorption of electrolytes and water**
II.	**Renal corpuscle**	**B.**	**Filtration of blood**
III.	**Distal part of collecting duct**	**C.**	**Impermeable to electrolytes**
IV.	**Descending loop of Henle**	**D.**	**Permeable to urea**

(1) I-A, II-B, III-D, IV-C

(2) I-C, II-D, III-B, IV-A

(3) I-B, II-A, III-D, IV-C

(4) I-C, II-D, III-A, IV-B

14. **Occurrence of excess urea in blood due to kidney failure is.**

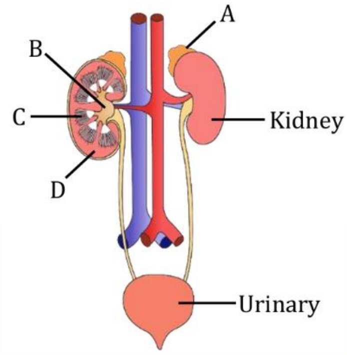
(1) Uremia

(2) Only cosuria

(3) Anuria

(4) Ketonuria

15. **Figure shows human urinary system with structures labelled A to D. Select option which correctly identifies them and gives their characteristics and/or functions: -**



(1) D-Cortex - outer part of kidney and do not contain any part of nephrons

(2) A-Adrenal gland - located at the anterior part of kidney. Secrete Catecholamines which stimulate glycogen breakdown

(3) B-Pelvis - broad funnel shaped space inner to hilum, directly connected to loops of Henle

(4) C-Medulla-inner zone of kidney and contains complete nephrons

16. **Which of the following statement is correct regarding excretion?**

(1) Descending limb of loop of Henle is permeable for electrolyte

(2) Glomerular filtration process is occurs in collecting duct

(3) Urea is more toxic nature compare to NH3

(4) Ascending limb of loop of Henle is impermeable to water.

17. **Find the correctly matched pair regarding kidneys.**

(1) Renal columns - Extensions of cortex, into the medullary pyramids.

(2) Renal pelvis - Cup shaped projections into which apex of medullary pyramids open.

(3) Renal cortex - Arranged as numerous pyramidal masses.

(4) Hilum - A notch at the concave inner margin of kidney.

18. **Given below are two statements:**

Assertion : The pH and ionic balance of blood is maintained by DCT and collecting duct.

Reason : DCTs of many nephrons open into a collecting ducts

Choose the most appropriate answer

(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) If Assertion is true but Reason is false

(4) If both Assertion and Reason are false

19. **During micturition, all of the given events occur, except**

(1) Contraction of skeletal muscles of urinary bladder

(2) Relaxation of urethral sphincter

(3) Release of urine

(4) Stretching of the urinary bladder initiate neural signals

20. **Angiotensin-II is responsible for all of the following, except**

(1) Vasoconstriction

(2) Increase in blood pressure

(3) Release of mineralocorticoids

(4) Decreased reabsorption of Na+ from renal tubules

21. **Path of urine drainage in the human excretory system is**

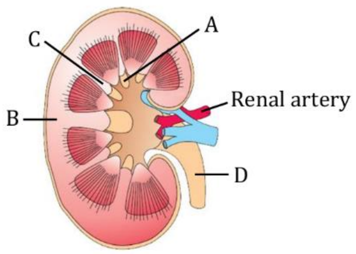
(1) Calyx - Collecting duct - Renal pelvis - Ureter Urinary bladder

(2) Renal pelvis - Collecting duct - Calyx - Ureter Urinary bladder

(3) Collecting duct - Calyces - Renal pelvis - Ureter - Urinary bladder

(4) Renal pelvis - Calyx - Collecting duct - Ureter Urinary bladder

22. **Identify A to D in the given structure and choose the correct option accordingly.**



(1) A-Calyx, B-Cortex, C-Renal column, D-Ureter

(2) A-Calyx, B-Cortex, C-Renal column, D-Urethra

(3) A-Urethra, B-Cortex, C-Renal column, D-Calyx

(4) A-Urethra, B-Calyx, C-Renal column, D-Cortex

23. **Given below are two statements:**

Statement-I: The Malpighian corpuscle, PCT and DCT of the nephron are situated in the medullary region of the kidney.

Statement-II: Vasa recta is absent or highly reduced in cortical nephrons.

In the light of the above statements, choose the most appropriate answer from option given below

(1) Statement I and Statement II both are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

24. **Which of the following is incorrect about proximal convoluted tubule (PCT)?**

(1) It is lined with simple cuboidal brush bordered epithelium.

(2) All essential nutrients, 40-50% electrolytes and water are reabsorbed here.

(3) It helps in maintenance of pH of the body fluid by selective secretion of H+ ion and absorption of HCO3−.

(4) It helps in maintenance of ionic balance of body fluid.

25. **Which of the following statements is correct?**

(1) Vasa recta is well developed in cortical nephron

(2) PCT and DCT are situated in the medulla of kidney

(3) The glomerulus encloses the Bowman's capsule

(4) The ascending limb of the Henle's loop extents as the DCT

26. **In human kidneys, the cortex extends in between the medullary pyramids as renal columns called**

(1) Hilum

(2) Calyces

(3) Columns of Bertini

(4) Capsule

27. **Selective secretion of H+ and K+ occurs in**

(1) PCT and Henle’s loop

(2) DCT and Henle’s loop

(3) Collecting duct and DCT

(4) Collecting duct and Henle’s loop

28. **Match List-I with List-II to find out the correct option.**

List-I	List-II
I. Length of kidney	**A. 10-12 cm**
II. Width of kidney	**B. 5-7 cm**
III. Thickness of kidney	**C. 120-170 gram**
IV. Weight of kidney	**D. 2-3 cm**

(1) I-A, II-B, III-D, IV-C

(2) I-C, II-D, III-B, IV-A

(3) I-B, II-A, III-D, IV-C

(4) I-C, II-D, III-A, IV-B

29. **All of the following organisms excrete their nitrogenous wastes in the form of pellet or paste with a minimum loss of water, except**

(1) Calotes

(2) Columba

(3) Corvus

(4) Canis

30. The characteristics common to urea, uric acid and ammonia is/are.

(i) They are nitrogenous wastes

(ii) they all need very large amount of water for excretion

(iii)They are all equally toxic

(iv)They are produced in the kidneys

(1) i, ii and iv

(2) i only

(3) i and iii

(4) i and iv

31. Which statement is correct regarding the haemodialysis procedure?

(1) Haemodialysis is the ultimate method in the correction of acute renal failure.

(2) Blood is drained from a convenient artery and pumped into dialysing unit containing heparin.

(3) The dialysing unit have a coiled paraffin tube surrounded by dialysing fluid.

(4) The composition of dialysing fluid is different from that of plasma.

32. Juxtaglomerular apparatus is a special sensitive region formed by cellular modifications in the;

(1) DCT and efferent arteriole at the point of their contact.

(2) DCT and afferent arteriole at the point of their contact.

(3) PCT and afferent arteriole at the point of their contact.

(4) PCT and efferent arteriole at the point of their contact.

33. High osmolarity gradient towards the inner medullary interstitial fluid in kidney is mainly due to

(1) NaCl and Water

(2) Water and Urea

(3) NaCl and Urea

(4) Water and Glucose

34. Which of the following statements are incorrect?

(a) Reabsorption of water occurs passively in the initial segment of nephron.

(b) Nitrogenous waste is absorbed by active transport.

(c) Conditional reabsorption of Na+ and water takes place in proximal convoluted tubule (PCT).

(d) Substances like glucose, amino acids, Na+, etc., in the filtrate are reabsorbed actively.

(1) (a) and (b)

(2) (b) and (c)

(3) (a) and (d)

(4) None of these

35. Which one of the following options shows a correct matching pair?

(1) Man-ureotelic

(2) Bird-Ammoniotelic

(3) Fish-Uricotelic

(4) Frog-uricotelic

36. Given below are two statements:

Assertion : In cortical nephrons, vasa recta is absent or highly reduced.

Reason : Cortical nephrons are mainly concern with concentration of urine.

Choose the most appropriate answer

(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) If Assertion is true but Reason is false

(4) If both Assertion and Reason are false

37. In humans, facultative reabsorption of water and Na+ occurs in

(1) Proximal convoluted tubule

(2) Descending limb of loop of Henle

(3) Ascending limb of loop of Henle

(4) Distal convoluted tubule

38. Select the total number of correct statements about the loop of Henle.

(a) Descending limb is permeable to water.

(b) Descending limb almost impermeable to electrolyte.

(c) Ascending limb impermeable to water.

(d) It allows transport of electrolyte only actively.

(e) At the tip of loop of Henle concentration of filtrate is 1200 mOsmL–1.

(f) It helps in maintaining high osmolarity in medullary interstitium.

(1) Six

(2) Three

(3) Four

(4) Five

39. What are columns of Bertini (Or Renal columns)

(1)Extensions of medulla in cortex

(2)Extensions of cortex in pelvis

(3)Extensions of medulla in pelvis

(4)extensions of cortex in medulla

40. All of the following hormones can increase the blood pressure in humans, except

(1) ADH

(2) Epinephrine

(3) ANF

(4) Aldosterone

41. The glomerular capillary blood pressure causes filtration of blood through three layers in a sequence of;

(1) Endothelium → Basement membrane → Epithelium of Bowman's capsule

(2) Epithelium of Bowman's capsule → Endothelium → Basement membrane

(3) Basement membrane → Endothelium → Epithelium of Bowman's capsule

(4) Epithelium of Bowman's capsule → Basement membrane → Endothelium

42. The first step in the urine formation is the filtration of the blood, which is carried by the _____(a)_____ and is called _____(b)_____. On an average _____(c)_____ mL of blood is filtered by kidneys per minute, which constitutes _____(d)_____ of the blood pumped out by each ventricle of the heart in a minute.

(1) (a) glomerulus; (b) filtration; (c) 800–900; (d) one-fourth

(2) (a) glomerulus; (b) filtration; (c) 1100–1200; (d) one-fifth

(3) (a) glomerulus; (b) filtration; (c) 1100–1300; (d) one-sixth

(4) (a) glomerulus; (b) filtration; (c) 2100–2500; (d) one-fifth

43. Read the following statements and choose the correct option.

Statement (A): Tubular epithelial cells in different segments of nephron perform reabsorption either by active or passive mechanism.

Statement (B): Nitrogenous wastes are absorbed by passive transport.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

44. ADH affects the kidney function by all of the following mechanisms, except

(1) Facilitates reabsorption of water from DCT

(2) It has constrictory effects on the blood vessels.

(3) It can increase blood pressure and thereby GFR

(4) Acts as a check on the renin-angiotensin mechanism by inhibiting it

45. Diameter of the renal afferent vessel is

(1) Same as that of efferent

(2) Smaller than that of efferent

(3) Larger than that of efferent

(4) There is no efferent vessel

46. The counter-current mechanism helps to maintain a concentration gradient. This gradient helps in;

(1) easy passage of water from medulla and thereby concentrating urine.

(2) easy passage of water from collecting tubule and thereby concentrating urine.

(3) easy passage of water from medullary interstitial fluid to collecting tubule and thereby diluting urine.

(4) inhibition of passage of water between the collecting tubule and medulla and so isotonic urine is formed.

47. Which factor(s) help(s) in maintaining an increasing osmolarity towards the inner medullary interstitium?

a. Counter-current pattern in vasa recta

b. Counter-current pattern in Henle's loop

c. Proximity between the Henle's loop and vasa recta

(1) a & b

(2) a, b & c

(3) c only

(4) a & c

48. Given below are two statements:

Assertion(A): Human kidneys can produce urine nearly four times concentrated than the initial filtrate formed.

Reason(R): On an average, 25-30 gm of urea is excreted out per day.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

49. Given below are two statements:

Statement-I: Dialysing fluid having the same composition as that of plasma including the nitrogenous wastes.

Statement-II: Haemodialysis is a boon for thousands of uremic patients all over the world.

(1) Statement I and Statement II both are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

50. The high osmolarity of the renal medulla is maintained by all of the following except;

(i) Active transport of salt from the upper region of the ascending limb.

(ii) The spatial arrangement of juxta medullary nephrons.

(iii) Diffusion of urea from the collecting duct.

(iv) Diffusion of salts from the ascending limb of the loop of Henle.

(v) Diffusion of salts from the descending limb of the loop of Henle.

(1) Only iv

(2) Only v

(3) ii & iii

(4) iv & v

Answer Key

1. (3)
2. (3)
3. (1)
4. (3)
5. (2)
6. (4)
7. (2)
8. (4)
9. (4)
10. (2)
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